

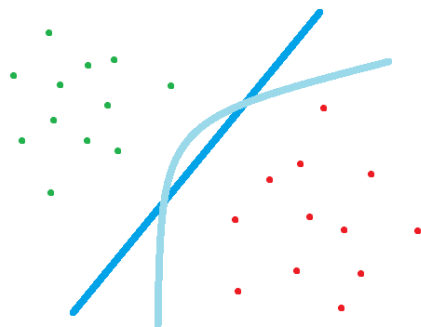
5. Bayes decision theory

Machine learning

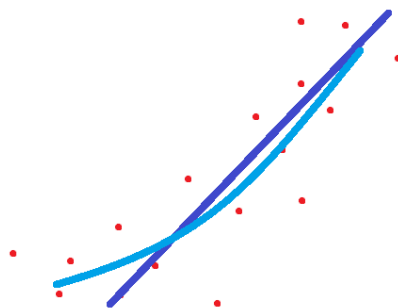
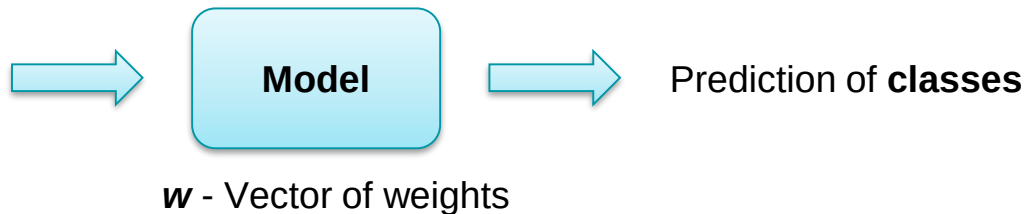
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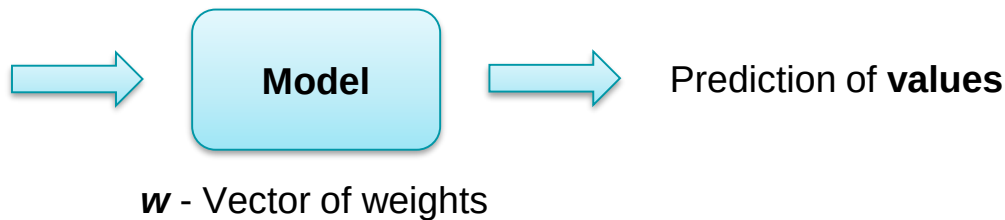
Linear / non-linear classifiers and regressors



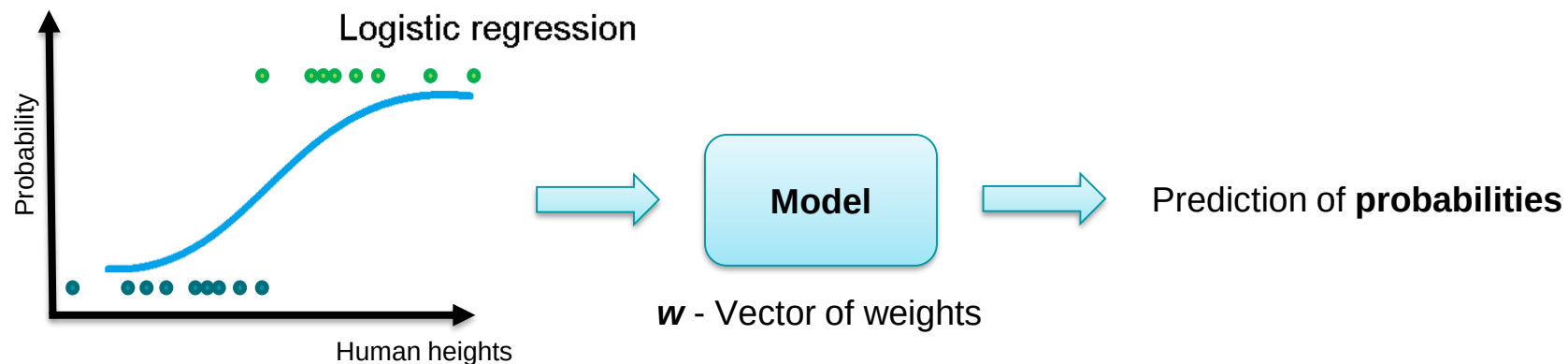
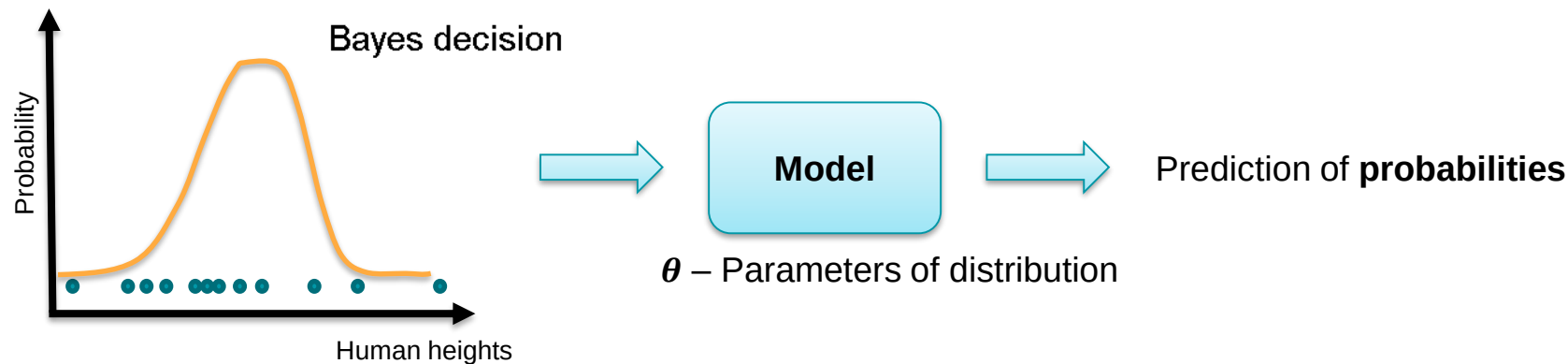
Linear / non-linear classification



Linear / non-linear regression



How are these systems different from others?



ML based on Bayes' rule

Methods:

- **Maximum likelihood (MLE)**
- **Maximum a posteriori probability (MAP)**
- **Bayesian inference**
- **Bayesian classifier**
 - Naive classifier
 - Non-naive classifier
- **Gaussian mixture model (GMM)**
- *Bayesian network*
- *Graphical models*
- *Bayesian optimization*

Introduction

Random variable – random variable is a variable whose value is uncertain

Probability – is a measure quantifying the likelihood that events will occur. The probability of an event A is written as:

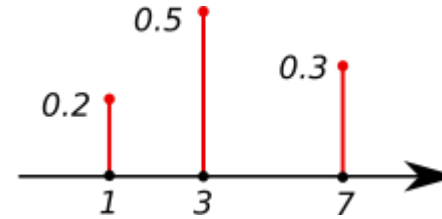
$$P(A), p(A) \text{ or } \Pr(A)$$

Probability distribution (generally) – provides the probabilities of occurrence of different possible outcomes in an experiment. They are generally divided into two classes: a **Probability mass function** (for a discrete distribution) and a continuous probability distribution is described by a **Probability density function**

Introduction

Probability mass function (PMF) – $f_X(x)$

$$P(X = x) = f_X(x)$$

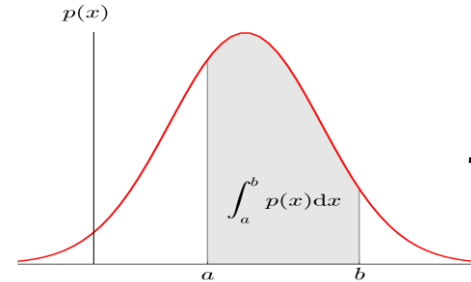


$$\sum_x f_X(x) = 1$$

Probability density function (PDF) – $f_X(x)$

$$P(X = x) = f_X(x) = 0$$

$$P(x_1 \leq X \leq x_2) = \int_{x_1}^{x_2} f_X(x) dx$$



$$\int_{-\infty}^{\infty} f_X(x) dx = 1$$

$$p_X(x) = P(X = x) \quad \text{or} \quad p_X(x) = P(x_1 \leq X \leq x_2)$$

Introduction

Observed data

Feature table

<i>Subject</i>	<i>Feature 1</i>	<i>Feature 2</i>	<i>Feature 3</i>	<i>Decision</i>
01	1	1	121.2	Positive
02	0	3	112.3	Negative
03	0	2	137.5	Positive
...	...	...	105.7	...
N	1	3	149.9	Positive

Contingency table

	<i>Blue</i>	<i>Green</i>	<i>Brown</i>	<i>Total</i>
Male	53	47	45	145
Female	65	33	82	180
Total	118	80	127	325

From observation of discrete variables (Feature table), we can derive any Contingency table, but not the other way around.

Can I do contingency table from a continuous random variable?

Introduction

Parametrical described PD - The shape of probability distribution is defined by mathematics describing a theoretical problem with several parameters. It can be used for both continuous or discrete variables.

Normal PD is defined by a mean value μ and a standard deviation σ

$$f(x|\mu, \sigma) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x - \mu)^2}{2\sigma^2}\right)$$

For example, an average male height in Czech republic is $\mu = 177,8 \text{ cm}$ with standard deviation of $\sigma = 10,6 \text{ cm}$. Assuming a normal distribution of human heights, we can obtain a PDF function from these known parameters.

Generally, we have a vector θ of parameters, e.g. for 1D normal distribution $\theta = [\mu, \sigma]$.

Introduction

Non-parametric described PF – only for a **discrete** distribution and it is described directly by probabilities of each discrete variable value.

Examples

$$x \in \{1, 3, 7\}$$

$$P(X = 1) = 0.1$$

$$P(X = 3) = 0.4$$

$$P(X = 7) = 0.5$$

$$x \in \{0.1, 1.5, 2.3, 26.7\}$$

$$P(X = 0.1) = 0.10$$

$$P(X = 1.5) = 0.25$$

$$P(X = 2.3) = 0.38$$

$$P(X = 26.7) = 0.27$$

$$x \in \{0, 1\}$$

$$y \in \{0, 1\}$$

$$P(X = 0, Y = 0) = 0.1$$

$$P(X = 1, Y = 0) = 0.2$$

$$P(X = 0, Y = 1) = 0.4$$

$$P(X = 1, Y = 1) = 0.3$$

Introduction

Example for a Bernoulli distribution

Parametric

$$f(x|\theta) = \theta^x(1 - \theta)^{1-x}$$

$$x \in \{0, 1\}$$

$$\theta = 0.4$$

Non-parametric

$$P(X = 0) = 0.6$$

$$P(X = 1) = 0.4$$

Feature table

<i>Subject</i>	<i>Feature 1</i>
01	1
02	0
03	0
...	...
N	1

If we have available observed data of a discrete random variable, we can obtain both parametric or non-parametric PD. But, for continuous variables, there is only parametric way of the notice.

Introduction

Joint probability - is the probability of two events (X,Y) happening together.

Joint probability distribution - is the joint probability function of two or more random variables.

Discrete case (mass function)

$$P(X = x, Y = y) = P(X = x \text{ and } Y = y) = P(X = x \cap Y = y)$$

For independent variables

$$P(X = x, Y = y) = P(X = x) \cdot P(Y = y)$$

Abbreviated notations

$$p_{X,Y}(x, y) = p_X(x) \cdot p_Y(y)$$

$$p(x, y) = p(x) \cdot p(y)$$

For dependent variables

$$P(X = x, Y = y) = P(X = x) \cdot P(Y = y|X = x)$$

$$p(x, y) = p(x|y) \cdot p(y)$$

Introduction

Continuous case (density function)

$$P(x_1 \leq X \leq x_2, y_1 \leq Y \leq y_2) = P(x_1 \leq X \leq x_2 \text{ and } y_1 \leq Y \leq y_2)$$

For independent variables

$$p_{X,Y}(x, y) = p_X(x) \cdot p_Y(y)$$

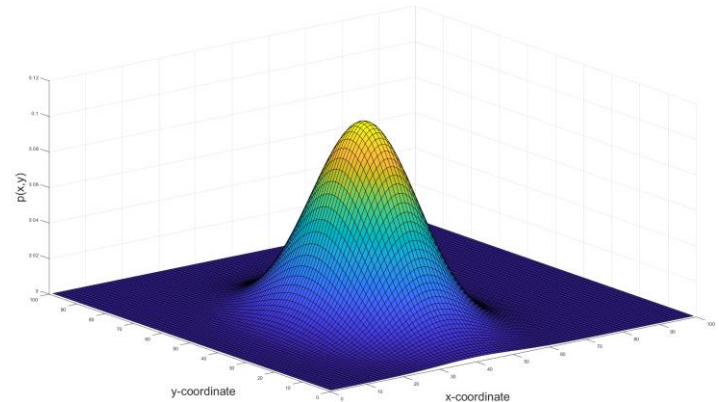
$$\text{resp. } p(x, y) = p(x) \cdot p(y)$$

For dependent variables

$$p_{X,Y}(x, y) = p_X(x) \cdot p_Y(y|x)$$

The following applies to a joint probability distribution:

$$p(x, y) = p(y, x)$$



Introduction

Marginal probability – is defined as a sum of a joint probability function of two or more random variables along specific dimensions. Thus, some variables are discarded (so called *marginalized out*) and this process is named *marginalizing*.

Marginalizing along Y random variable

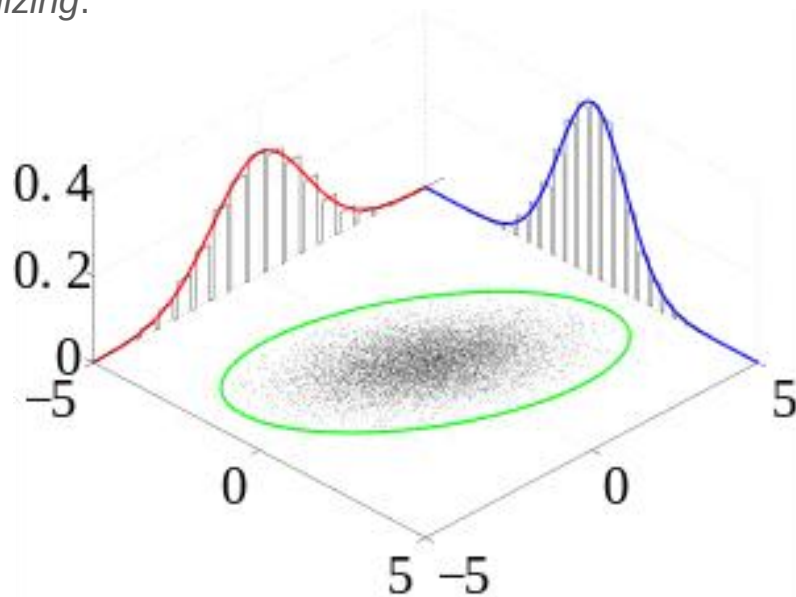
$$P(X = x) = \sum_y P(X = x, Y = y)$$

Marginalizing along X random variable

$$P(Y = y) = \sum_x P(X = x, Y = y)$$

Abbreviated notation

$$p(x) = \sum_y p(x, y)$$



Introduction

Conditional probability – is a value of probability, that an event B will occur if another event A has occurred. The concept of conditional probability is one of the most fundamental and one of the most important in probability theory.

$$P(X|Y = y)$$

Kolmogorov definition: conditional probability of events A and B is given a rate of probability that both events A and B occur and marginal probability of event B.

$$p(x|y) := \frac{p(x, y)}{p(y)}$$

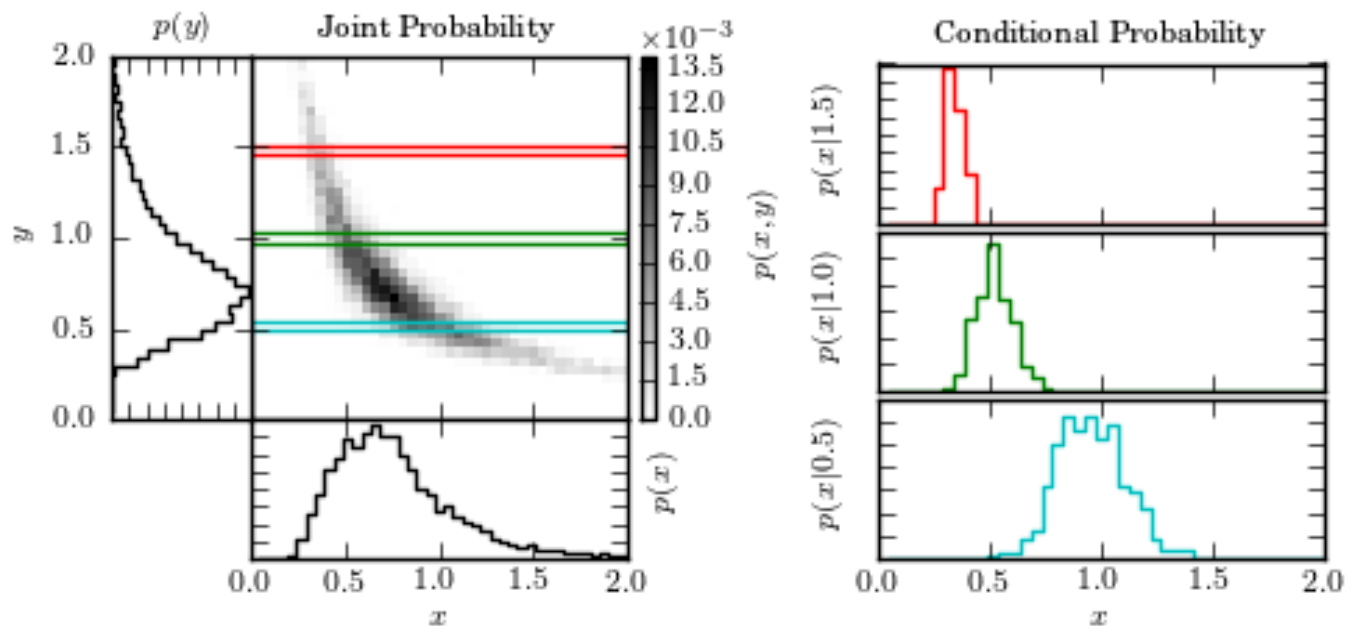
Then
$$p(x, y) = p(x|y) \cdot p(y)$$

In contrast with joint probability, the conditional probability of event A given B is not equal to the probability of event B given A.

$$p(x|y) \neq p(y|x)$$

$$P(A | X = X(w)) = P(A | X)(w)$$

Introduction



Introduction - example

X – shape

Y – color

$$P(X = \text{circle}) = p_X(\text{circle})$$

$$P(X = \text{square}) = p_X(\text{square})$$

$$P(Y = \text{blue}) = p_Y(\text{blue})$$

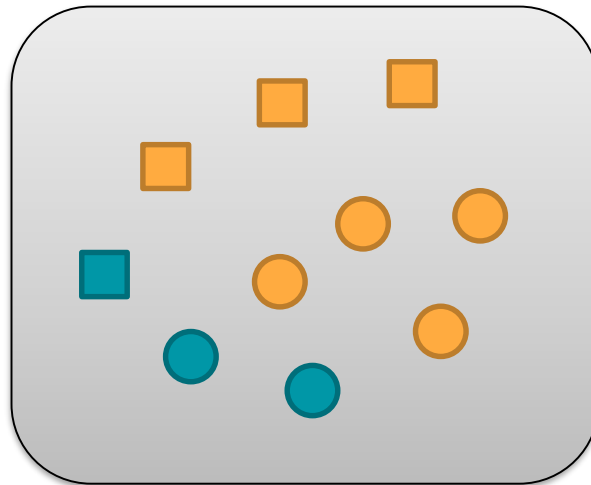
$$P(Y = \text{orange}) = p_Y(\text{orange})$$

$$p(\text{blue}, \text{square})$$

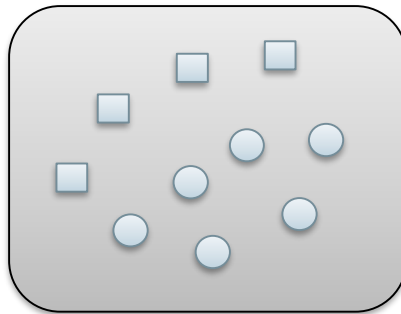
$$p(\text{blue}|\text{square})$$

$$p(\text{blue}|\text{square}) \neq p(\text{square}|\text{blue})$$

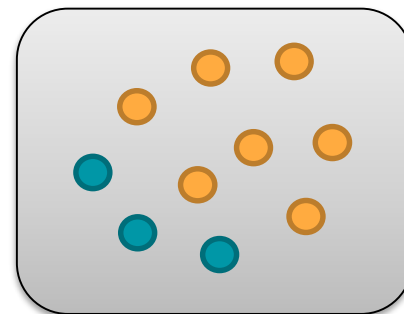
$$p(\text{blue}, \text{square}) = p(\text{square}, \text{blue})$$



X – shape



Y – color



Bayes decision theory

Bayes' rule

Law of total probability describes a marginal probability of event X via a conditional probability of given event Y and marginal probability.

From definition of conditional probability:

$$p(x|y) := \frac{p(x, y)}{p(y)} \implies p(x, y) = p(x|y)p(y)$$

$$p(x) = \sum_y p(x, y) \implies p(x) = \sum_{i=1}^n p(x|y_i)p(y_i)$$

Total probability

Alternatives:

$$p(x, y) = p(x|y)p(y)$$

$$p(y, x) = p(y|x)p(x)$$

$$p(x, y) = p(y, x)$$

$$p(x|y)p(y) = p(y|x)p(x)$$

Based on this equation, we can derive **Bayes' rule**

$$p(x|y) = \frac{p(y|x)p(x)}{p(y)} = \frac{p(y|x)p(x)}{\sum_{i=1}^n p(y|x_i)p(x_i)} = \frac{p(y|x)p(x)}{\sum_{i=1}^n p(y|x_i)p(x_i)}$$

Bayes' terminology

Aprior probability $p(x)$

probability is the probability an event will happen before you take any new evidence into account (what we know before performing the experiment)

Likelihood $p(y|x)$

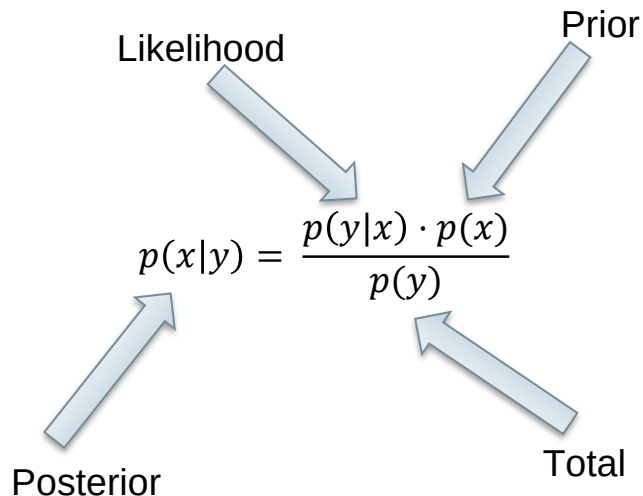
quantifies the likelihood that the observed data would have been observed (it can be used to rank the plausibility of model parameters X)

Posterior probability $p(x|y)$

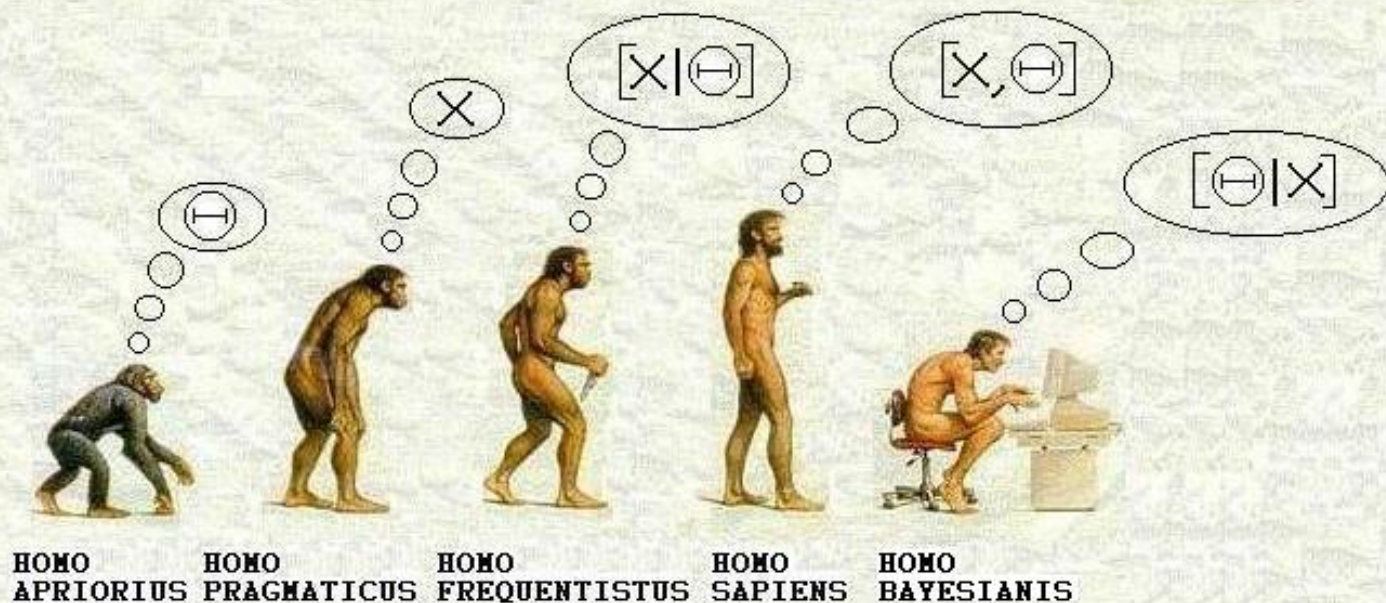
is the probability an event will happen after all evidence has been taken into account

Total $p(y) = \sum_{i=1}^n p(y|x_i)p(x_i)$

marginal likelihood; this is the total probability of observing the evidence



(YET ANOTHER) HISTORY OF LIFE AS WE KNOW IT...



Example 1

Bayes' probability

A ... a person has a heart attack

B_1 ... a person is non – smoker

B_2 ... a person is light smoker

B_3 ... a person is heavy smoker

Measurement:

	B_1	B_2	B_3	Total
A	15	40	45	100

Probability:

$$p(B_1|A) = 0,15$$

$$p(B_3|A) = 0,45$$

Prior:

$$p(B_1) = 0,40$$

$$p(B_3) = 0,25$$

$$p(A) = 0,05$$

Postprior:

$$p(A|B_1) = ???$$

$$p(A|B_3) = ???$$

Calculus:

$$p(A|B_1) = \frac{p(B_1|A)p(A)}{p(B_1)} = \frac{0,15 \cdot 0,05}{0,4} = 0,019$$

$$p(A|B_3) = \frac{p(B_3|A)p(A)}{p(B_3)} = \frac{0,45 \cdot 0,05}{0,25} = 0,09$$

Example 2

Bayes' probability – prevalence

A ... disease occurs

B₊ ... positive result of test with sensitivity of 98%

Case 1: If I have a positive test, but I feel healthy. What probability is that I really have a disease?

by conventional statistic: 98%

by Bayes' probability:

we have available a new findings about prevalence of disease – 10%

$$p(A|B_+) = \frac{p(B_+|A)p(A)}{p(B_+)} = \frac{0,98 \cdot 0,1}{0,116} = 84\%$$

$$p(B_+) = p(B_+, A) + p(B_+, \bar{A}) = p(B_+|A)p(A) + p(B_+|\bar{A})p(\bar{A}) = 0,98 \cdot 0,1 + 0,02 \cdot 0,9 = 0,116$$

$$p(y) = \sum_{i=1}^n p(y|x_i)p(x_i)$$

Example 3

Bayes' probability – probability distribution

A ...is normal distribution $\mathcal{N}(\mu, \sigma^2)$

B ...is set of observation of human height $B = \{x_1, x_2, \dots, x_n\}$

We assume a normal distribution of human heights.

*The task is to find parameters of posterior distribution A,
i. e. mean $\hat{\mu}$ and variance $\hat{\sigma}^2$, which will best represents measured data B.*

Solution:

Methods on next slides (MLE, MAP, ...)

MLR based on Bayes' rule

Methods:

- Maximum likelihood (MLE)
- Maximum a posteriori probability (MAP)
- Bayesian inference
- Bayesian classifier
 - Naive classifier
 - Non-naive classifier
- Gaussian mixture model
- Bayesian network
- Graphical models
- Bayesian optimization

MLR based on Bayes' rule

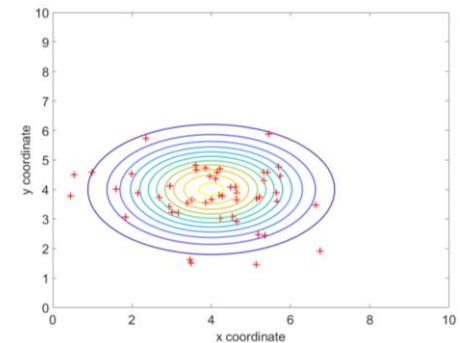
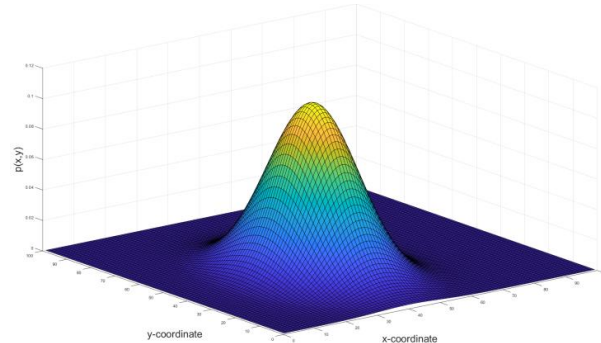
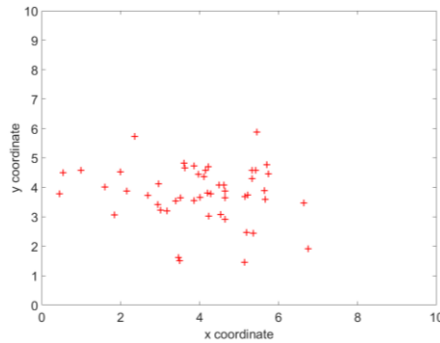
Utilization:

- Decision systems with prior knowledge
 - Expert decision systems in medicine (support decision making in the diagnosis process, correction of test results)
 - To find good strategies for solving tasks in a domain with uncertainty
 - Open-source or commercial software
- Classification task based on selected features
 - Support disease classification in the diagnose process
 - Signal processing, Image classification
 - Natural language processing
 - Genetic analysis, Phylogenetic trees
- For machine learning algorithm
 - Optimization of hyper-parameters (e.g. learning rate)

Maximum likelihood

Maximum likelihood (ML, MLE)

The estimation method that finds the parameters of a probability distribution based on maximizing a **likelihood** function for the set of observed data $x = \{x_1, x_2, \dots, x_n\}$. Thus, the task is to find argument likelihood function, its value is maximal. The resulting model **is the best for generating the observed data**.



The main advantage is an analytic solution to the problem (if exist) and thus easy to find the parameters of the preselected distribution. Nevertheless, it can be solved also any optimization technique (brute-force, iterative algorithm).

Definition of ML

Task:

to find the θ that maximizes likelihood:

$$\theta^* = \operatorname{argmax}_{\theta} p(\theta|x) = \operatorname{argmax}_{\theta} p(x|\theta)$$

modification:

$$\theta^* = \operatorname{argmin}_{\theta} -\log p(x|\theta)$$

if we know that $x_1, x_2, \dots, x_n \sim \mathcal{N}(\theta)$, where $\theta = [\mu, \sigma^2]$

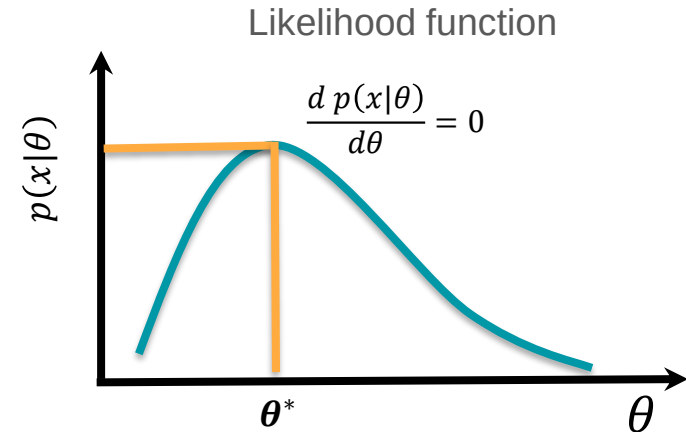
$$\mu^* = \operatorname{argmin}_{\mu} -\log p(x|\mu, \sigma^2)$$

$$\sigma^{2*} = \operatorname{argmin}_{\sigma^2} -\log p(x|\mu, \sigma^2)$$

$$p(\theta|x) = \frac{p(x|\theta)\cancel{p(\theta)}}{\cancel{p(x)}}$$

we assume a uniform distribution of probability

for finding the minimum of function is irrelevant



Solution – for Normal PD

$$p(A|B) = \frac{p(B|A)p(A)}{p(B)}$$

PDF of normal distribution,

$$f(x|\mu, \sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right)$$

PDF for n **independent** random samples (the likelihood)

$$p(x_1, \dots, x_n|\mu, \sigma^2) = \prod_{i=1}^n p(x_i, |\mu, \sigma^2) = \left(\frac{1}{2\pi\sigma^2}\right)^{\frac{n}{2}} \exp\left(-\frac{\sum_{i=1}^n (x_i - \mu)^2}{2\sigma^2}\right)$$

$$x_i \sim \mathcal{N}(\mu, \sigma^2) \text{ i.i.d.}$$

$$\mathcal{L}(\mu, \sigma^2) = -\log p(\mathbf{x}|\boldsymbol{\theta}) = -\log p(x_1, \dots, x_n|\mu, \sigma^2)$$

Solution:

$$\mu^* = \operatorname{argmin}_{\mu} \mathcal{L}(\mu, \sigma^2) = \frac{1}{n} \sum_{i=1}^n x_i = \bar{x}$$

$$\sigma^{2*} = \operatorname{argmin}_{\sigma^2} L(\mu, \sigma^2) = \frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})^2 = \hat{\sigma}^2$$

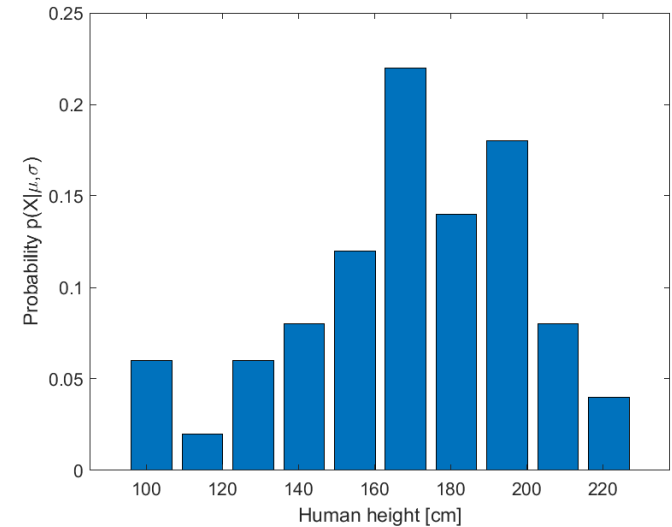
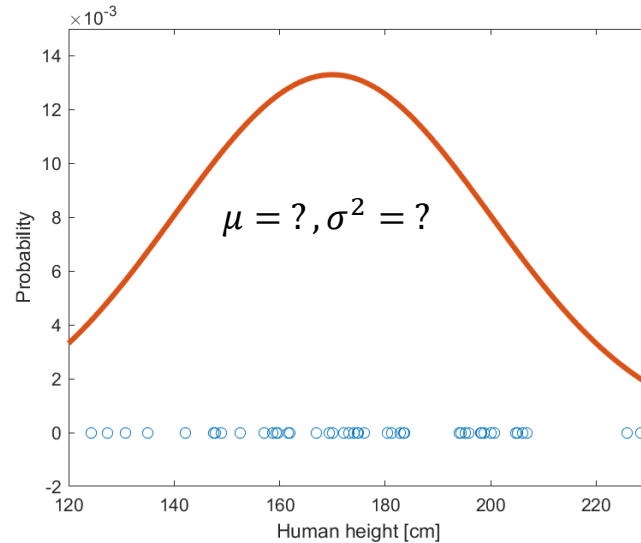
Can be proved:

$$E[\hat{\sigma}^2] = \frac{n-1}{n} \sigma^2$$

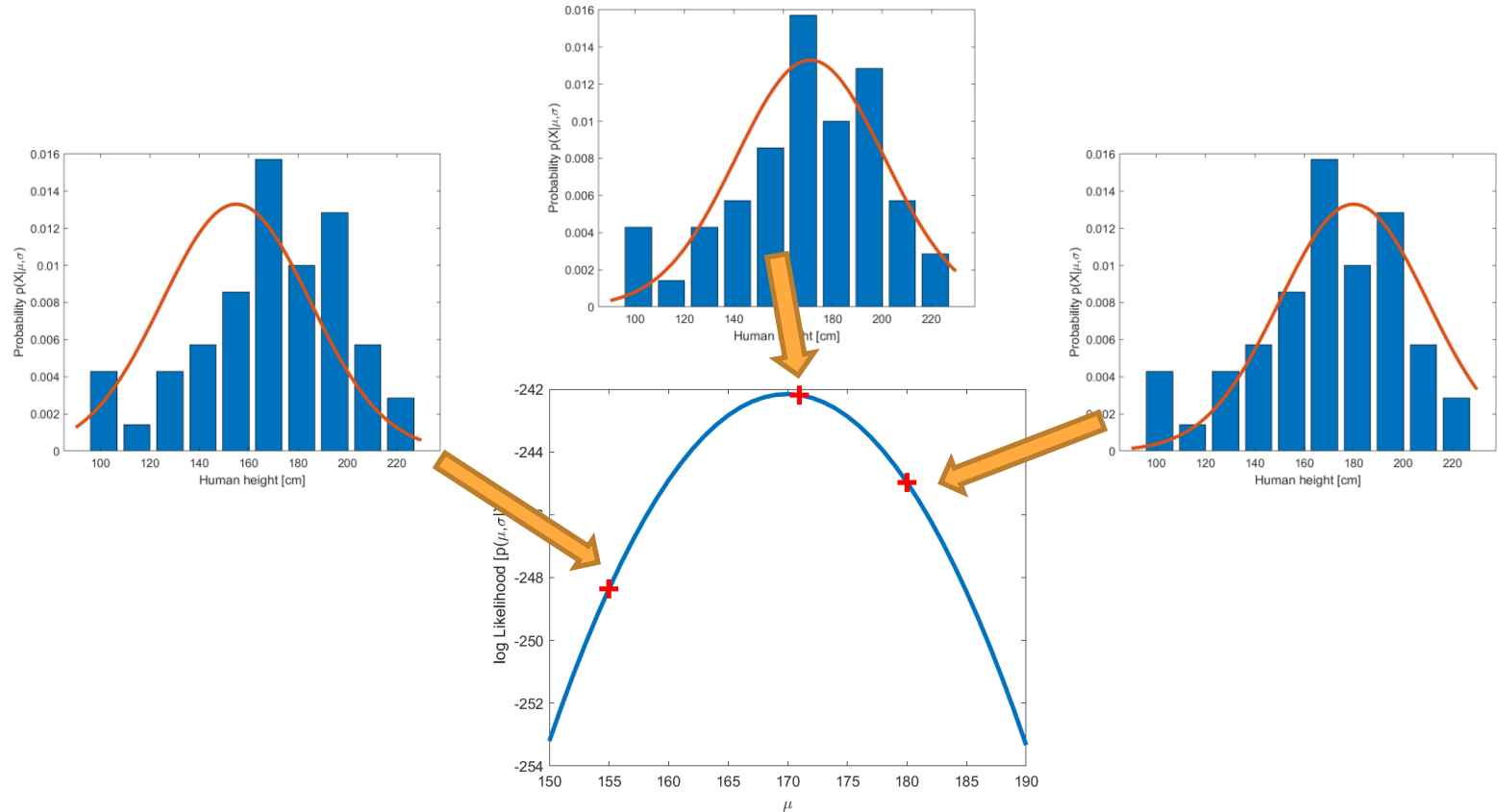
Example

We have N measurements of human height $X = x_1, x_2, \dots, x_N \sim \mathcal{N}(\mu, \sigma^2)$

$x_1 = 172 \text{ cm}$
 $x_2 = 189 \text{ cm}$
 $x_3 = 168 \text{ cm}$
 $\vdots$
 $x_N = 163 \text{ cm}$



Likelihood function



Summary

Pros:

- Easy to compute
- Intuitive interpretation
- Asymptotic properties
 - unbiased – with more data estimate converges in the mean to the true value $\lim_{N \rightarrow \infty} E[\hat{\theta}_{ML}] = \theta_0$
 - consistent – with more data estimate converges in:
 - probability $\xrightarrow{\text{orange arrow}} \lim_{N \rightarrow \infty} \text{prob}\{\|\hat{\theta}_{ML} - \theta_0\| \leq \epsilon\} = 1, \text{ where } \epsilon \text{ is arbitrarily small}$
 - mean squared error $\xrightarrow{\text{orange arrow}} \lim_{N \rightarrow \infty} E[\|\hat{\theta}_{ML} - \theta_0\|^2] = 0$
 - efficient – with more data estimate have the lowest possible variance that any estimate can achieved (Cramer-Rao lower bound)
 - with more data estimate converges to Normal distribution with mean θ_0 (central limit theorem)
- Invariant under reparameterization

Cons:

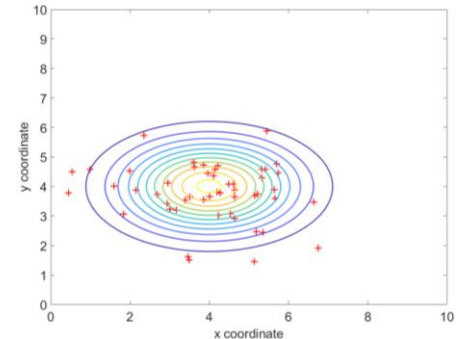
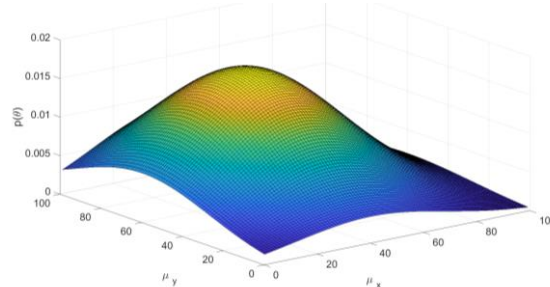
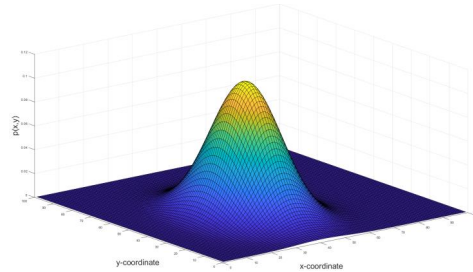
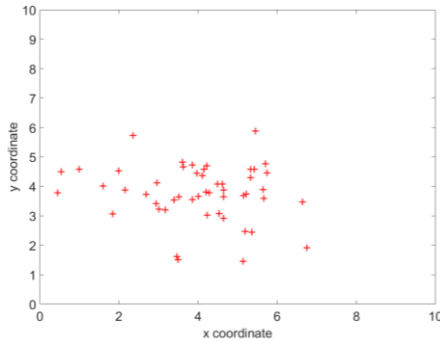
- Point estimate – no representation of uncertainty (approximation based on data – likelihood)
- Not suitable for the multi-modal likelihood PDF
 - the highest mode may not correspond to the wanted maximum of the likelihood function
- Overfitting
- Black swan theory – zero probability problem – e.g. Logarithm of zero
- Existence & uniqueness of ML is not guaranteed

Maximum a-posteriori probability

Maximum a posteriori probability (MAP)

The estimation method that find the parameters of a probability distribution based on maximizing a **a-posterior probability** $p(x, \theta)$ for the set of observed data $x = \{x_1, x_2, \dots, x_n\}$.

The resulting model, which is defined by estimated parameters, **is the most probable model for given (observed) data** x taking into account a prior distribution of individual models.



Definition

Maximum Likelihood

$$\theta_{MAP}^* = \operatorname{argmax}_{\theta} p(\theta|x) = \operatorname{argmax}_{\theta} p(x|\theta)$$

$$p(A|B) = \frac{p(B|A) \cdot p(A)}{p(B)}$$

Maximum a posteriori probability

$$\theta_{MAP}^* = \operatorname{argmax}_{\theta} p(\theta|x) = \operatorname{argmax}_{\theta} \frac{p(x|\theta) \cdot p(\theta)}{\cancel{p(x)}}$$

$$\theta_{MAP}^* = \operatorname{argmax}_{\theta} p(x_1, \dots, x_n|\theta) \cdot p(\theta)$$

- regularization of ML estimation
- using a prior distribution of a related event
- the posterior distribution $p(\theta|x)$ can be calculated using Bayes' theorem from likelihood function $p(x|\theta)$ and the prior $p(\theta)$

Conjugate prior

If the posterior distribution $p(\theta|x)$ and the prior $p(\theta)$, are from the same probability distribution family (only exponential) then they are called conjugate distributions and the prior is called conjugate prior for the likelihood function $p(x|\theta)$.

$$p(\theta|x) = p(x|\theta) \cdot p(\theta)$$

If we want to ensure the same distribution of the posterior $p(\theta|x)$ and a prior $p(\theta)$, we need to choose an appropriate conjugate distributions for a likelihood distribution $p(x|\theta)$.

Examples:

- normal distribution of the likelihood function needs normal distribution for a prior,
- Bernoulli distribution needs beta distribution,
- more at Table of conjugate distributions: https://en.wikipedia.org/wiki/Conjugate_prior

WHY?

- analytic solution of the product $p(x|\theta)$ and $p(\theta)$ will be less difficult than with another distribution
- more transparent expression of a prior distribution update (by likelihood function)

Solution – for Normal PD

Bayes' theorem

$$p(A|B) = \frac{p(B|A)p(A)}{p(B)}$$

PDF of Normal distribution

$$f(x|\mu, \sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right)$$

likelihood function PDF for n **independent** random samples

$$p(x_1, \dots, x_n|\mu, \sigma^2) = \prod_{i=1}^n p(x_i|\mu, \sigma^2) = \left(\frac{1}{\sqrt{2\pi\sigma^2}}\right)^n \exp\left(-\frac{\sum_{i=1}^n (x_i - \mu)^2}{2\sigma^2}\right)$$

$x_i \sim \mathcal{N}(\mu, \sigma^2)$ i. i. d.

prior PDF according to **conjugate prior** for Normal distribution with known σ of likelihood fcn.

$$p(\mu|\mu_P, \sigma_P^2) = \frac{1}{\sqrt{2\pi\sigma_P^2}} \cdot \exp\left(-\frac{(\mu - \mu_P)^2}{2\sigma_P^2}\right)$$

$\mu \sim \mathcal{N}(\mu_P, \sigma_P^2)$

$\mu_P, \sigma_P^2, \sigma^2$ – constants

$$\mathcal{P}(\mu; \sigma^2, \mu_P, \sigma_P^2) = -\log [p(x_1, \dots, x_n, |\mu, \sigma^2) \cdot p(\mu|\mu_P, \sigma_P^2)]$$

Solution:

$$\mu_{MAP}^* = \operatorname{argmin}_{\mu} \mathcal{P}(\mu; \sigma^2, \mu_P, \sigma_P^2) = \frac{n\sigma_P^2}{n\sigma_P^2 + \sigma^2} \cdot \bar{x} + \frac{\sigma^2}{n\sigma_P^2 + \sigma^2} \cdot \mu_P$$

weights $\swarrow$ $\searrow$ convex combination of $\bar{y}$ and μ
 $\nwarrow$ sample mean (ML) $\swarrow$ prior mean

Summary

Pros:

- Analytic solution is usually possible when conjugate priors are used (for exponential family distributions only)
 - otherwise any numeric optimization must be used (gradient methods, Newton's method, Monte Carlo simulation, EM algorithm)
- Easy to compute, interpretable
- Avoid overfitting – regularization
- Tends to look like ML asymptotically

Cons:

- The highest mode may not correspond to the wanted maximum of the posterior probability
- Point estimate – no representation of uncertainty in θ (posterior distribution)
- Not invariant under reparameterization
- Must assume a prior on θ

Example

ML/MAP – blood pressure

A ... is normal distribution of measurement $\mathcal{N}(\mu, \sigma^2)$

B ... is set of measurement of blood pressure from one subject $B = \{x_1, x_2, \dots, x_n\}$

We assume a normal distribution of a blood pressure measurement.

The task is to find parameters of the posterior distribution $A - p(A, B)$

i.e. mean $\hat{\mu}$ and variance $\hat{\sigma}^2$, which will best represents measured data B .

In this example we consider known σ as a standard deviation of used sphygmomanometer – estimation is simplified to the estimation of mean $\hat{\mu}$ only.

ML Solution:

$$\mu_{ML}^* = \operatorname{argmin}_{\mu} \mathcal{L}(\mu; \sigma^2) = \operatorname{argmin}_{\mu} -\log p(B|\mu; \sigma^2) = \bar{x}$$

MAP Solution:

For MAP estimation we need to know a prior distribution of a related event

C – conjugate distribution with $\mathcal{N}(\mu_P, \sigma_P^2)$, with known μ_0 – mean value of blood pressure in population and σ_P^2 – standard deviation in population.

$$\begin{aligned} \mu_{MAP}^* &= \operatorname{argmin}_{\mu} \mathcal{P}(\mu; \sigma^2, \mu_P, \sigma_P) = \operatorname{argmin}_{\mu} -\log [p(B|\mu, \sigma^2) \cdot p(\mu|\mu_P, \sigma_P^2)] = \\ &= \frac{n\sigma_P^2}{n\sigma_P^2 + \sigma^2} \cdot \bar{x} + \frac{\sigma^2}{n\sigma_P^2 + \sigma^2} \cdot \mu_P \end{aligned}$$

Example

ML / MAP – blood pressure

	x_1	x_2	x_3
Observed	121	109	115

	μ	σ
Prior	120	12
Posterior	?	6

ML Solution:

$$\mu_{ML}^* = \bar{x} = \frac{121 + 109 + 115}{3} = 115 \text{ mmHg}$$

MAP Solution:

$$\mu_{MAP}^* = \frac{n\sigma_p^2}{n\sigma_p^2 + \sigma^2} \cdot \bar{x} + \frac{\sigma^2}{n\sigma_p^2 + \sigma^2} \cdot \mu_0 = \frac{3 \cdot 12^2}{3 \cdot 12^2 + 6^2} \cdot 115 + \frac{6^2}{3 \cdot 12^2 + 6^2} \cdot 120 = 116 \text{ mmHg}$$

Summary

Easy to compute and intuitively interpretable often solved using an analytic solution

Maximum likelihood

- What parameters θ of model will be the best for generating the observed data x ?
- ML estimation attempts to find the parameter vector θ that maximizes $p(\theta, x)$ – likelihood, thus the model that the best generates observed data.
- The estimation is based only on the observed data x .
- Overfitting

Maximum a posteriori probability

- What parameters θ of model the best explains the observed data x ?
- MAP estimation attempts to find the parameter vector θ that maximizes $p(x, \theta)$ – posterior, thus the most probable model (parameters of distribution) assuming observed data x .
- The estimation requires a prior $p(\theta)$, which favors/disadvantages of individual distributions.
- It has a regularization in the form a prior (overfitting elimination).

Bayesian inference

Bayesian inference

- Assume a prior dist. $p(\theta)$ and a total probability $p(x)$
- Estimates whole posterior PD – not point estimate only
- Bayesian procedures – minimizes a loss as average error over all θ
- Subjective – belief-based priors as information from previous studies or experts
- Objective – non-informative priors based on the observed data

Pros:

- Directly answer questions (What is the probability of something (θ in some interval)?)
- Avoid overfitting – usage of an appropriate prior
- Automatic model selection based on the evidence (“Occom's razor” – complex model might not be better than simple one)
- Admissible – better method does not exist for all solutions
- We know an uncertainty of estimate

Cons:

- Must assume a prior
- Exact computation can be intractable (integrating)

Priors:

- Non-informative – small influence of estimation by a prior – all is about data
- Robust – e.g. with heavy tails – if we are not very confident in our prior
- Conjugate or mixture of conjugate – easy to compute

Bayesian classifiers

Bayesian classifiers

Given a random variables $\mathbf{x}$ and set of the class $y = [y_1, \dots, y_K]$. We want obtain the posterior probability of the class membership:

$$p(y_k | \mathbf{x})$$

We apply a Bayesian rule:

$$p(y_k | \mathbf{x}) = \frac{p(\mathbf{x} | y_k) \cdot p(y_k)}{p(\mathbf{x})}$$

where K is number of classes and. Generally, variables $\mathbf{x}$ is a vector of d features, thus defines a dimension of space.

$$predict = \underset{y_k}{\operatorname{argmax}} p(\mathbf{x} | y_k) \cdot p(y_k) = \underset{y_k}{\operatorname{argmax}} p_{\theta}(x_1, \dots, x_d | y_k) \cdot p(y_k)$$



This conditional probability for k -class can be derived from observed data.



This probability can be derived from observed data as frequency of samples with k -class.

We select from set $\mathbf{x}$ only samples with k -th class and fit parameters of probability distribution $p(x_1, \dots, x_d | y_k)$ on this data by e.g. ML estimation.

However, this distribution is d -dimensional and therefore fitting can be difficult. It is possible only for a simple distributions e.g. Gaussian distribution.

Naive Bayes classifiers

Naive Bayes classifiers

$$predict = \operatorname{argmax}_{y_k} p(\mathbf{x} | y_k) \cdot p(y_k) = \operatorname{argmax}_{y_k} p(x_1, \dots, x_d | y_k) \cdot p(y_k)$$

This probability can be derived from observed data as frequency of samples with k-class.



Assume that the features are mutually independent, conditional on given category y_k

$$predict = \operatorname{argmax}_{y_k} p(x_1, \dots, x_d | y_k) \cdot p(y_k) = \operatorname{argmax}_{y_k} p(y_k) \cdot \prod_{i=1}^d p_{\theta}(x_i | y_k)$$

The task is to find distributions $p(x_i | y_k)$ described by parameter vector $\theta_{i,k}$ for i -th feature conditional by k -th class, thus we obtain $k \times d$ distributions.

The parameters of this distribution can be found with probability model based on observed data as follows.

We have observed set of n samples for i -th feature $[f_1, f_2, \dots, f_n]$. Then we estimate parameter $\theta_{i,k}$ for set of samples $\mathbf{f}_{i,k}$ corresponding to k -class. We are fitting distribution only to samples corresponding to k -class, because $p(x_i | y_k)$ is conditioned by y_k .

$$\theta_{MAP}^* = \operatorname{argmax}_{\theta_{i,k}} p(\theta_{i,k} | \mathbf{f}_{i,k}) = \operatorname{argmax}_{\theta_{i,k}} \frac{p(\mathbf{f}_{i,k} | \theta_{i,k}) \cdot p(\theta_{i,k})}{p(\mathbf{f}_{i,k})} \quad \mathbf{f}_{i,k} = \mathbf{f}(x_i | y_k)$$

Naive Bayesian classifiers

- Naive – assume that features are mutually independent:
 - We get less complex model – easy to estimate, but...
 - in the real-world data it is not true – features are often somehow mutually dependent, but...
 - even the wrong and simple model can still perform better than the correct and complex model.

Pros:

- Robust to missing features – only existing features can be used in the product
- Robust to dependency of features
- Selection of the most appropriate distribution for each feature
- Parameters of distributions can be estimated by ML/MAP – easy to compute and interpretable
- Because of conditional independency of features on given category we can estimate θ more accurately with less data – product of 1D distributions instead of complex ND distribution

Cons:

- Must assume distributions of θ

Gaussian mixture model

Gaussian mixture model estimation

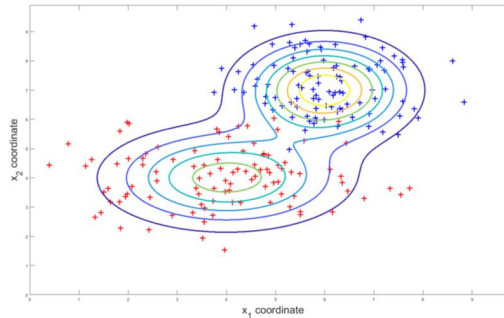
Unsupervised learning method for **unlabeled** data clustering into **known number** of classes
(in general for whatever probability distribution)

In general, the main task is fitting a set of N-dimensional probability distributions on N-dimensional observed data.

But, we don't know anything about the populations a priori (except you believe that they are Gaussian)

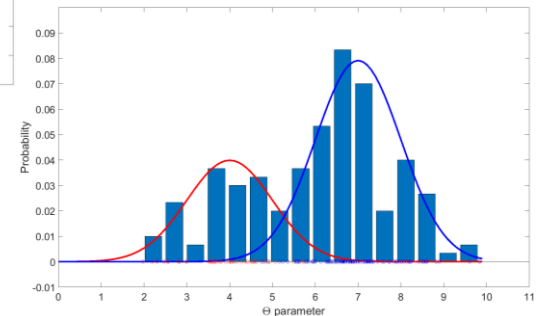
A probabilistic model that assumes all the data points are generated from a **mixture** of a finite number of Gaussian distributions with unknown parameters.

The aim is thus to **estimate parameters** of Gaussian distributions.



2D clustering into classes

1D fitting on distribution



Gaussian function

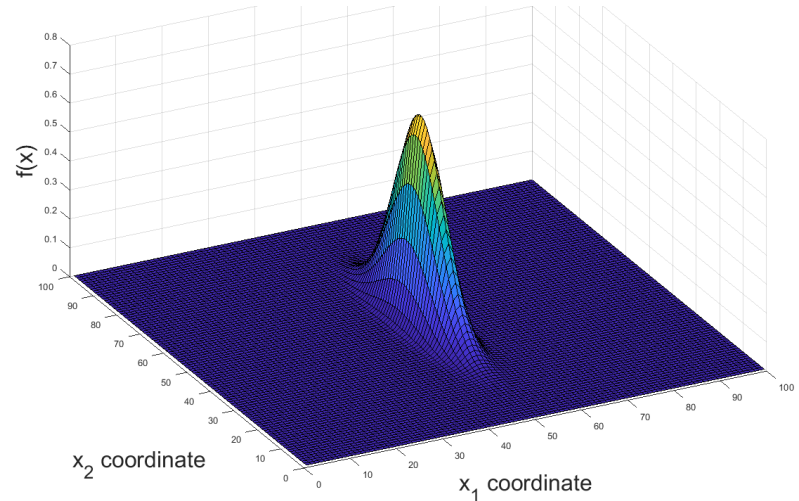
d-dimensional Gaussian function:

$$f(\mathbf{x}) = (2\pi)^{-\frac{d}{2}} |\mathbf{\Sigma}|^{-\frac{1}{2}} \exp\left(-\frac{1}{2}(\mathbf{x} - \boldsymbol{\mu})^T \mathbf{\Sigma}^{-1}(\mathbf{x} - \boldsymbol{\mu})\right)$$

$$\begin{aligned} \boldsymbol{\mu} &= (\mu_1, \mu_2, \dots, \mu_d)^T \\ \mathbf{x} &= (x_1, x_2, \dots, x_d)^T \\ \mathbf{\Sigma} &= \begin{bmatrix} \sigma_1^2 & \cdots & \sigma_1\sigma_d \\ \vdots & \ddots & \vdots \\ \sigma_d\sigma_1 & \cdots & \sigma_d^2 \end{bmatrix} \end{aligned}$$

for 2-dimensional case:

$$\mathbf{x} = \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} \quad \boldsymbol{\mu} = \begin{pmatrix} \mu_1 \\ \mu_2 \end{pmatrix} \quad \mathbf{\Sigma} = \begin{bmatrix} \sigma_1^2 & \sigma_1\sigma_2 \\ \sigma_2\sigma_1 & \sigma_2^2 \end{bmatrix}$$



Mixture model

for one parametric model (K-modal distr.)

$$f(x; a_1, a_2, \dots, a_K) = \sum_{k=1}^K w_k p(x; a_k)$$

for two parametric model (K-modal distr.)

$$f(x; a_1, a_2, \dots, a_K, b_1, b_2, \dots, b_K) = \sum_{k=1}^K w_k p(x; a_k, b_k) \quad \sum_{k=1}^K w_k = 1$$

weights can be a-priori (no learned) or a-posteriori

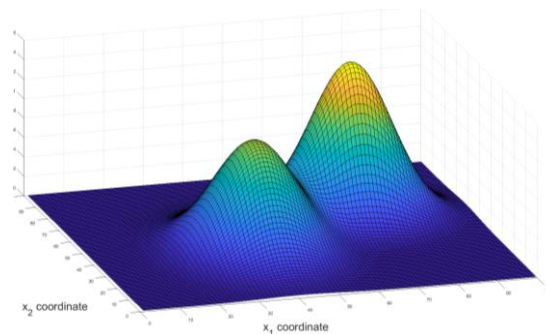
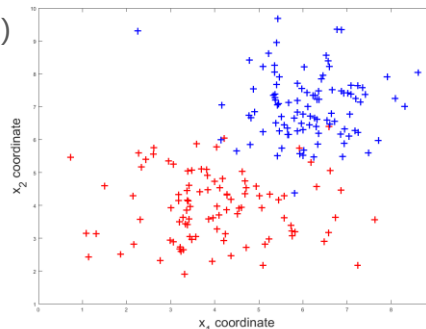
for 2D Gaussian mixture model (bimodal distribution K=2)

$$f(x; \mu_1, \sigma_1^2, \mu_2, \sigma_2^2) = \sum_{k=1}^2 w_k p(x; \mu_k, \sigma_k^2)$$

Alternatively for multivariate GMM

$$f(x; \mu_1, \mu_2, \Sigma_1, \Sigma_2) = \sum_{k=1}^2 w_k p(x; \mu_k, \Sigma_k)$$

$$f(x; \mu_1, \dots, \mu_K, \Sigma_1, \dots, \Sigma_K) = \sum_{k=1}^K w_k p(x; \mu_k, \Sigma_k)$$



Gaussian mixture model estimation

Problem definition for K-modal case:

$$f(\mathbf{x}|\boldsymbol{\mu}_1, \dots, \boldsymbol{\mu}_K, \boldsymbol{\Sigma}_1, \dots, \boldsymbol{\Sigma}_K, w_1, \dots, w_K) = \sum_{k=1}^K w_k p(\mathbf{x}|\boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k)$$

- to find the parameters (mean, variance and weights) of Mixture model

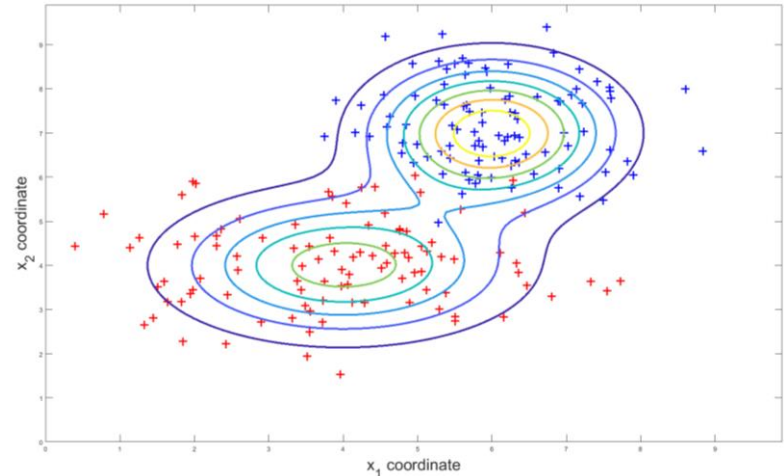
Bimodal 2D Gaussian mixture model estimation ($d=2, K=2$)

$$f(\mathbf{x}|\boldsymbol{\mu}_1, \boldsymbol{\mu}_2, \boldsymbol{\Sigma}_1, \boldsymbol{\Sigma}_2, w_1, w_2) = \sum_{k=1}^2 w_k p(\mathbf{x}|\boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k)$$

- solution via MLE, which seek to maximize the likelihood of the observed data given the model parameters, is unfortunately impossible in polynomial time (by analytical nor numerical methods)



Therefore, the estimation must be solved via iterative algorithms (e.g. EM, MCMC)



Expectation – maximization

Iterative method to find ML or MAP estimate of model parameters in two steps.

Let us the probability model, which is mixture (sum) of weighted Gaussians - with $d=2$ dimensions and $K=2$ Gaussians

$$P(\mathbf{x} = x_i; \boldsymbol{\Theta} = \boldsymbol{\Theta}_k) = p(x_i; \boldsymbol{\Theta}_k) \quad \boldsymbol{\Theta} = (\boldsymbol{\mu}_1, \boldsymbol{\mu}_2, \boldsymbol{\Sigma}_1, \boldsymbol{\Sigma}_2, w_1, w_2)$$

E – step (calculate probability)

$$p(x_i; \boldsymbol{\Theta}_k) = \frac{w_k p(x_i; \boldsymbol{\Theta}_k)}{\sum_{j=1}^K w_j p(x_i; \boldsymbol{\Theta}_j)}$$

M – step (fitting by MLE) For k -th Gaussian and N observations

$$w_k = \frac{1}{N} \sum_{i=1}^N p(x_i; \boldsymbol{\Theta}_k), \quad \mu_k = \frac{\sum_{i=1}^N p(x_i; \boldsymbol{\Theta}_k) x_i}{\sum_{i=1}^N p(x_i; \boldsymbol{\Theta}_k)}, \quad \boldsymbol{\Sigma}_k = \frac{\sum_{i=1}^N p(x_i; \boldsymbol{\Theta}_k) (x_i - \mu_k)(x_i - \mu_k)^T}{\sum_{i=1}^N p(x_i; \boldsymbol{\Theta}_k)}$$

- slow for big dimensionality
- and have a high percentage of missing data

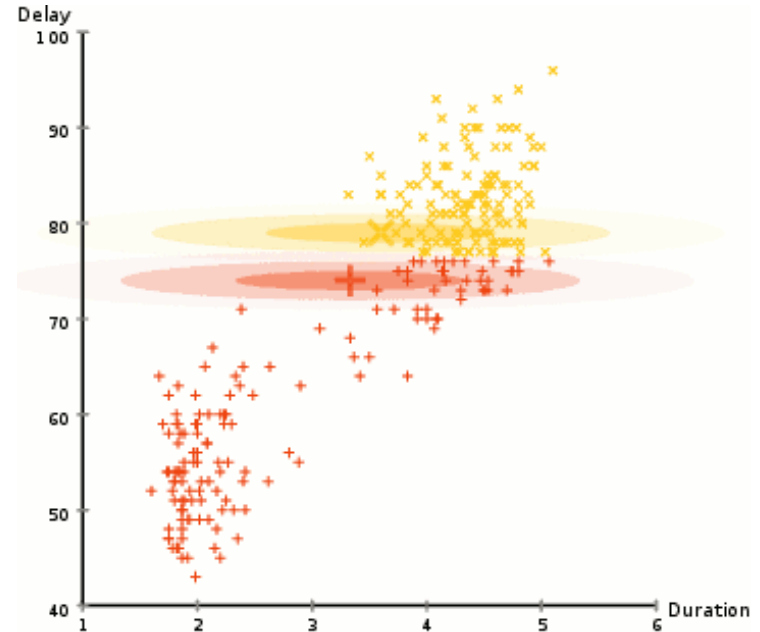
Gaussian mixture model estimation by EM

Algorithm for case of two 2D Gaussian (d=2, K=2)

1. Initialization of $\theta = (\mu_1, \mu_2, \Sigma_1, \Sigma_2, w_1, w_2)$
2. calculate probability (likelihood) $p(x_i; \theta_1)$ and $p(x_i; \theta_2)$ (E-step)
3. calculate w_1 and w_2 as mean of $p(x_i; \theta_k)$
4. re-estimate parameters μ_1, μ_2, Σ_1 and Σ_2
5. repeat step 2. - 4. until stopping criteria are not met

Stopping criterion:

- Maximal number of iterations
- Change of parameters is under limit $|\mu^{i+1} - \mu^i| < \epsilon$



Gaussian mixture model estimation

Unsupervised learning method for **unlabeled** data clustering into **known number** of classes
(in general for whatever probability distribution)

Examples:

- Clustering based on specific features assuming Gaussian distribution of them
- Support of disease classification
- Parameters estimation in diffusion measurement (sonography, dMRI, DWI/DTI/DKI)
- Image segmentation
- Audio analysis

In contrast with k-means, the Gaussian mixture models (GMM) clustering can handle even very oblong clusters and performs a soft assignment. In fact, we can derive the K-means algorithm as a particular limit of EM for Gaussian mixtures as follows.

Logistic regression

Logistic regression

Logistic regression is a method for creating the statistical model that is used in **classification problems**. It allows us to take some features and predict the correct class, or better - the **probability of that class**.

Linear *versus* logistic regression:

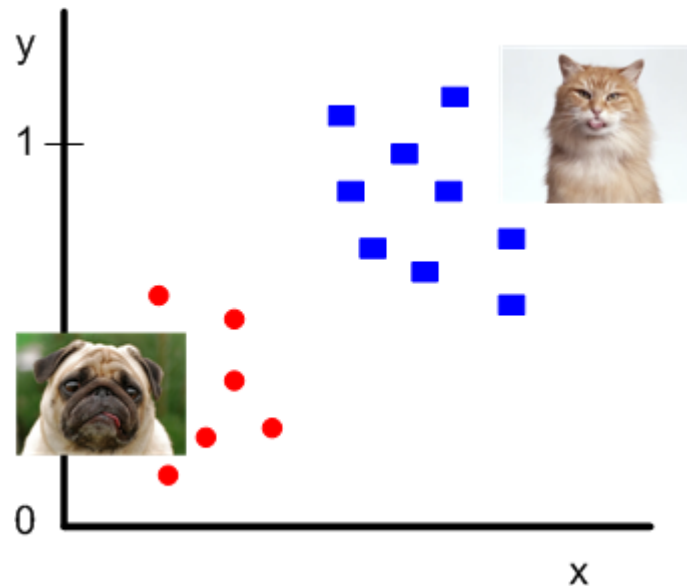
- we can predict if student pass the exam (yes/no) based on number of hours spent by studying. That is task for logistic regression.
- we can predict how many points the student get (“almost” continuous variable) based on number of hours spent by studying. That is task for linear regression.

Definition

We want to predict what is the probability that we classify CAT (i.e. $y_i=1$), based on given input (vector of features $\mathbf{x}_i$) and parameters of the model $\mathbf{w}$. Written in math:

$$p(y_i = 1 | \mathbf{x}_i, \mathbf{w}) = ?$$

It doesn't make sense to use the linear model, because it will just fit the line through our data. But we can use some function to transform this linear model into something more convenient.



Sigmoid function

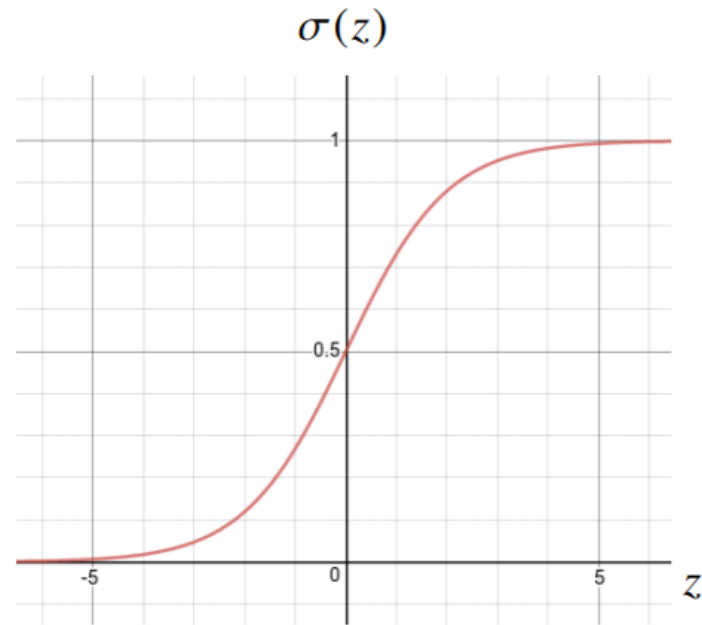
For this purpose we will use the sigmoid function:

$$\sigma(z) = \frac{1}{1 + e^{-z}}$$

The useful properties of this function are:

- whatever the input is, it is transformed between 0 and 1.
- the derivative of a sigmoid with respect to input is very nice/useful (see later):

$$\frac{d \sigma(z)}{d z} = \sigma(z) \cdot (1 - \sigma(z))$$



Logistic regression

Now, we will use this sigmoid function to transform our linear model:

$$\theta = \sigma(\mathbf{w}^T \mathbf{x}) = \frac{1}{1 + e^{-\mathbf{w}^T \mathbf{x}}}$$

Bernoulli probability distribution: $f(x|\theta) = \theta^x(1 - \theta)^{1-x}$

We can now look on our data in a different way: We can interpret each label as a Bernoulli random variable given the model described by sigma function.

Now, we define likelihood as a “probability” that observed data x come from specific model (Bernoulli) with parameters $\mathbf{w}$. This likelihood function for my independent and identically distributed random variable has the following form:

$$\mathcal{L}(\mathbf{w}|\mathbf{y}, \mathbf{x}_1, \dots, \mathbf{x}_n) = \prod_i^n \sigma(\mathbf{w}^T \mathbf{x}_i)^{y_i} (1 - \sigma(\mathbf{w}^T \mathbf{x}_i))^{1-y_i}$$

$$\operatorname{argmax}_{\mathbf{w}} L(\mathbf{w}|\mathbf{y}, \mathbf{x}_1, \dots, \mathbf{x}_n)$$

Solution

It turns out that maximizing likelihood function can be difficult due to multiplications. Better way is to maximize Log of likelihood function:

$$\mathcal{L}_{log} = \log \mathcal{L}(\mathbf{w} | \mathbf{y}, \mathbf{x}_1, \dots, \mathbf{x}_n) = \sum_i y_i \log(\sigma(\mathbf{w}^T \mathbf{x}_i)) + (1 - y_i) \log(1 - \sigma(\mathbf{w}^T \mathbf{x}_i))$$

We have to maximize this function with respect to $\mathbf{w}$. So we must take the derivative of $\mathcal{L}_{log}$

$$\frac{d \mathcal{L}_{log}(\mathbf{w} | \mathbf{y}, \mathbf{x}_1, \dots, \mathbf{x}_n)}{d w^{(j)}} = \sum_i (y_i - \sigma(\mathbf{w}^T \mathbf{x}_i)) x_i^{(j)}$$

And finally we can write the gradient ascent optimization (we will go “uphill”):

$$w_{new}^{(j)} = w_{old}^{(j)} + \mu \sum_i (y_i - \sigma(\mathbf{w}^T \mathbf{x}_i)) x_i^{(j)}$$

Bayes' theory

"An Essay towards solving a Problem in the Doctrine of Chances"

"The Theory that Would not Die"

